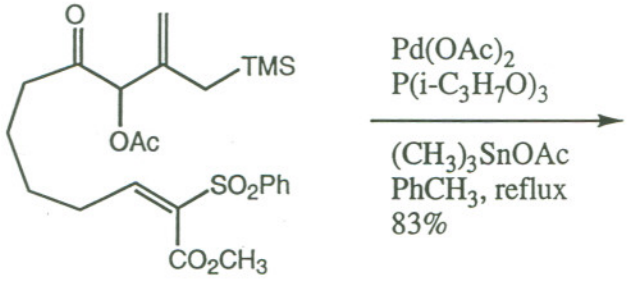
ORGANIC CHEMISTRY CUMULATIVE EXAM

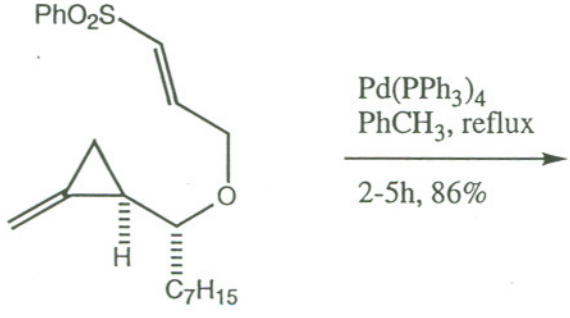
December 7, 1996

One of the principal uses of transition metal catalysts in organic synthesis is the stabilization of otherwise highly reactive intermediates (e.g. transition metal complexes of cyclobutadiene). Trimethylene methane (TMM) is just such a reactive intermediate, which can be formed by pyrolysis of methylenecyclopropane. The last two issues of *J. Am. Chem. Soc.* illustrate two synthetic applications of transition metal stabilization of TMMs. Draw the products, including stereochemistry, and the likely mechanism of the formation and reaction of the Pd stabilized TMMs in the following examples.

(1)



(2)



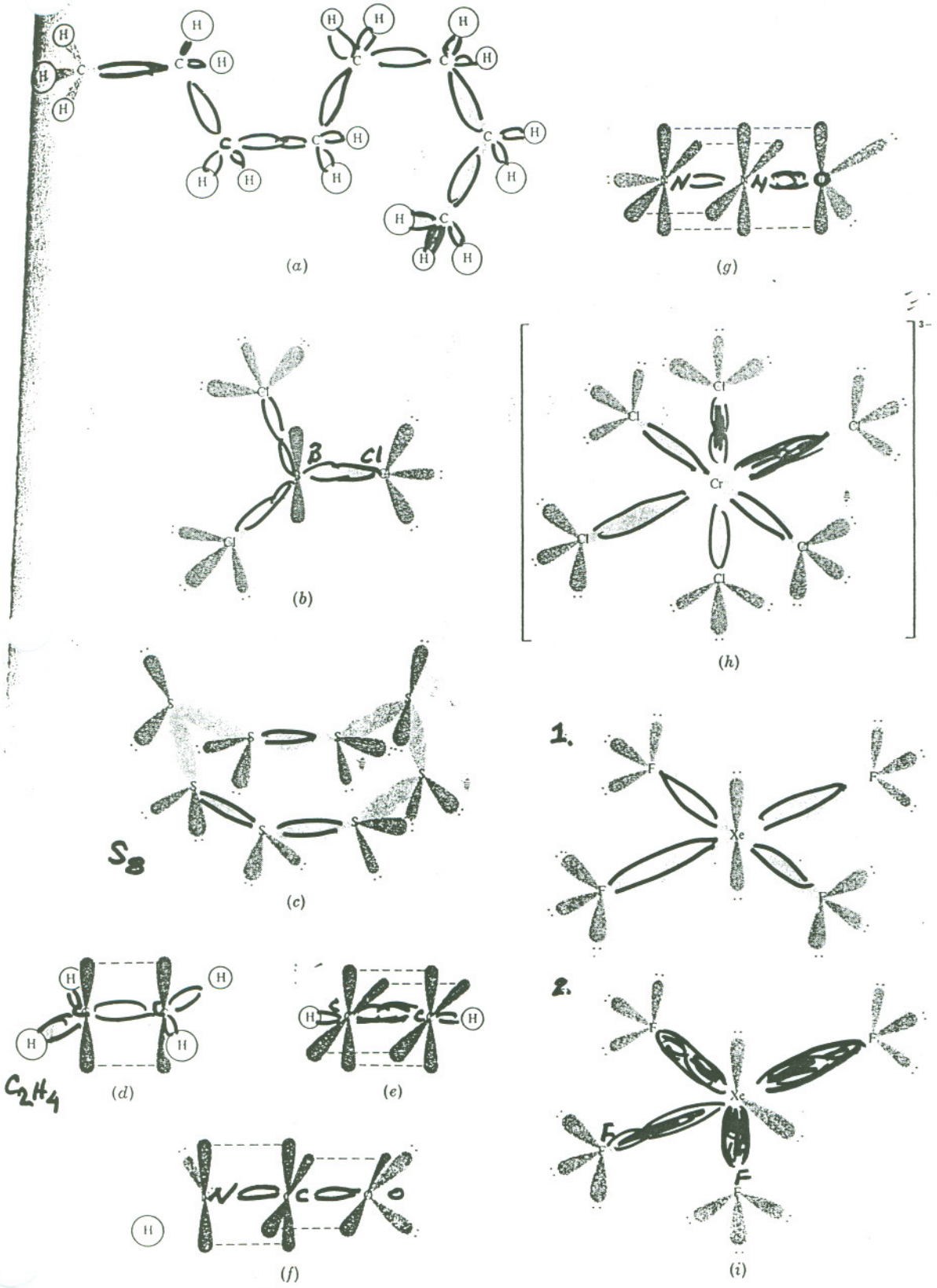
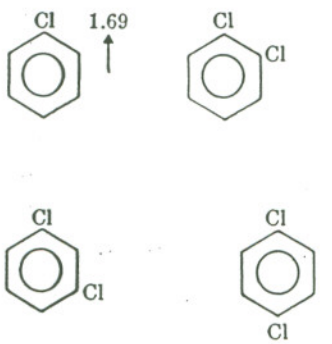


Fig. 15-17

2. If the dipole moment for a particular bond in a molecule is known, it is often possible to predict semiquantitatively the dipole moment for the molecule containing two such bonds by vectorial analysis. For a planar molecule, the dipole moment (μ) is the square root of the sum of the squares of the components in the x and y directions (i.e., $\mu = [(\mu_x)^2 + (\mu_y)^2]$. The dipole moment of chlorobenzene is 1.69 D. Predict the dipole meoment of o-, m- and p-dichlorobenze.